Module 1:
Introduction to Geometrical Optics

Course 1 of *Optical Engineering*: First Order Optical System Design

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Geometrical optics

"*Geometrical optics is either very simple, or else it is very complicated*"

Richard P. Feynman

What do we want to know?

1. Where is the image?
2. How large is it?
3. How bright is it?
4. What is the image quality?

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History

- 280BC Euclid's *Catoptrics* light goes in straight line. However light was emitted by eye (not)
- Law of *reflection* was correctly formulated in Euclid's book
- Hero of Alexandria, in his *Catoptrics* (first century BC), also maintained that light travels with infinite speed.
- Adopted the rule that light rays always travel between two points by the *shortest path*
- 1000 AD Arab philosopher “Alhazan” (Abu'ali al-hasan ibn al-haytham) first person to realize that light actually travels from the object seen to the eye
- 1676 Olaf Römer demonstrated that light must have a *finite* velocity, using his timings of the successive eclipses of the satellites of Jupiter
- 100-170AD Claudius Ptolemy rough law of *refraction* was studied experimentally (only close to normal incidence)
- 1621 Dutch mathematician Willebrord Snell comes up with correct law using sines. Does not publish
- 1637 French philosopher René Descartes was the first to publish, French countries call Snell’s law Descartes law of refraction
- 1658 French mathematician Pierre de Fermat demonstrated that all three of the laws of geometric optics can be accounted for on the assumption that light always travels between two points on the path which takes the *least time*
- 1861 Maxwell’s equations

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